The Arrow of Time as a Theorem: Transforming the Second Law of Thermodynamics from Law to Theorem on Reversible Dynamics

Why Reversible Mechanics Obeys the Second Law, and How Time-Symmetric Dynamics Yields a One-Way Law: Reproducible Processes Never Lower Entropy, and Every Non-Lowering Process Is Reproducible; No Return After a Rise; Reversal Returns Trajectories, Not Macrostates; Machine-Checked in Lean 4 and Mathlib

Mathematical Physics · Foundations of Thermodynamics

Time-symmetric dynamics, a one-way law

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Abstract. Since Loschmidt the arrow of time has posed one question: how can microscopic dynamics that run equally well backwards produce a law that runs one way? The Second Law of Thermodynamics has stood as a law, not a theorem, because every derivation from reversible mechanics needed a premise to close that gap. This paper transforms it into a theorem. Take any finite state space and let the microscopic dynamics be any permutation of it: reversible, information-preserving, Liouville's theorem in finite form. A macrostate is a set of microstates and its Boltzmann multiplicity W the number of them. A macroscopic process from A to B is reproducible when every microstate prepared in A is carried into B: the prepared record persists, which is existence stated on the record and Jaynes's reproducibility. The main theorem: a process is reproducible under some reversible dynamics if and only if $W(A) \leq W(B)$. The forward half is the Second Law, Jaynes's 1965 argument made exact: reversible mechanics obeys the Second Law on every reproducible process. The converse, achievability, says every pair of macrostates that does not lower entropy is reproduced by some permutation of the state space; dynamics with further conservation laws restrict which. The arrow follows as theorems. After a rise in W no reversible dynamics reproduces the return. Each direction is reproducible exactly when W is unchanged. And Loschmidt's objection is located: the inverse dynamics returns every trajectory it carried, yet whenever W rose it does not carry the macrostate back, because reversal returns trajectories, not macrostates. The one hypothesis is reproducibility, and it is where time's direction enters: a process is prepared, then observed, and the act of repeatable preparation supplies the arrow that the time-symmetric dynamics cannot. Existence is self-recursive: every state lies in its own record, and the record of anything in a record is that record. On the nine states $2^a 3^b$ graded by the prime count $\Omega = a + b$ the multiplicities are 1, 2, 3, 2, 1; the count is raised reproducibly from 0 to 1 and from 1 to 2, never lowered from 1 to 0, and the widest shell has no reproducible exit. Every theorem is proved in Lean 4 with Mathlib, each cone pinned. The arrow of time, as the Second Law states it for reproducible processes, is a theorem of reversible dynamics.

Keywords *second law of thermodynamics · Boltzmann entropy · reversible dynamics · reproducibility · Liouville theorem · Loschmidt paradox · arrow of time · prime degrees of freedom · formal verification · Lean 4 · Mathlib*

1. How to read this paper

Every theorem named here is a declaration of one file, `Second_Law_Theorem_v1_0_0_CC.lean`, written in Lean 4.19.0 against Mathlib v4.19.0 (de Moura and Ullrich 2021; Mathlib Community 2020). It declares no axiom and uses no `sorry`. Ten theorems, each cone printed and pinned by `#guard_msgs` at the foot of the file; nine rest on Lean's three standard axioms and one on propext and Quot.sound alone. The file compiles silent and is printed whole in Appendix A. Its design and non-negotiables are written into its header, so that any later editor reads them before touching a line.

1.1 The verdict

> Reversible mechanics obeys the Second Law. On every finite state space with reversible dynamics, a reproducible macroscopic pro-

cess never lowers the Boltzmann multiplicity, and every pair of macrostates that does not lower it is reproduced by some permutation of the state space. Time-symmetric dynamics yields a one-way law: after a rise the return is never reproducible, each direction is reproducible exactly when the multiplicity is unchanged, and the inverse dynamics undoes every trajectory and never the rise. The arrow enters through the one hypothesis, reproducibility, delivered by the act of repeatable preparation. Existence is self-recursive. On the prime count the multiplicities are 1, 2, 3, 2, 1, and the widest shell has no reproducible exit. The arrow of time, as the Second Law states it, is a theorem.

2. The arrow of time: from law to theorem

Clausius stated the Second Law in the 1850s, heat does not pass by itself from a colder to a hotter body, and named entropy in 1865 (Clausius 1865); Kelvin stated it for engines (Kelvin 1851). Boltzmann (1872) derived the rise of entropy from kinetic theory, on the assumption of molecular chaos; Loschmidt (1876) objected that reversed velocities give the same mechanics with entropy falling, and Zermelo (1896) that bounded mechanics recurs. Since then each derivation has carried a premise: a low-entropy past (Albert 2000; Lebowitz 1993), a canonical initial state (Jarzynski 1997; Crooks 1999), or axioms on adiabatic accessibility (Carathéodory 1909; Lieb and Yngvason 1999).

The puzzle is the arrow of time. The microscopic laws run equally well backwards, yet the Second Law runs one way. Jaynes (1965) found the form in which the one-way law follows from the two-way mechanics. Physics uses the law to say which macroscopic processes are possible, and a process is something an experimenter can do again: prepare the system in macrostate A, and it reaches macrostate B, every time. Under Liouville's theorem the prepared region keeps its volume, so if it lands inside B then B is at least as large as A: the Boltzmann entropy did not fall. This paper proves that argument exactly, and proves its converse, so the law becomes a characterization: the possible processes are precisely the ones that do not lower entropy.

3. The setting

The state space is any finite type, read as phase space at a fixed resolution. The microscopic dynamics is any permutation φ of it: reversible, with an exact inverse, and preserving counts, which is Liouville's theorem in finite form. A macrostate is a finite set of microstates, and its Boltzmann multiplicity is the number of its microstates (`W`); the Boltzmann entropy is $k \log W$.

A process from A to B is *reproducible* under φ when every microstate prepared in A is carried into B (`Reproducible`). This is existence, stated on the prepared record: the record "prepared in A" persists as the record "found in B". The condition is asked on the prepared microstates and nowhere else. That restriction is necessary, not cosmetic. If persistence of every record were demanded on every microstate, the pulled-back records of a reversible dynamics would keep their total fibre size fixed while every fibre could only grow, so every fibre would stay the same and no arrow could occur; reproducibility on the prepared record is exactly what leaves room for the arrow.

4. The theorem

Theorem 1 (the Second Law). If a process is reproducible under a reversible dynamics, the Boltzmann multiplicity does not fall: $W(A) \le W(B)$ (`second_law`). The dynamics is injective on A and sends A into B, so B has at least as many microstates as A.

Theorem 2 (achievability). If $W(A) \le W(B)$, some permutation of the state space reproduces the process (`achievability`). The dynamics is chosen after A and B; a given physical dynamics, with energy and other conserved quantities, can only restrict which pairs it reproduces. Choose a part of B with exactly as many microstates as A, match A to it one to one, and extend the matching to a permutation of the whole space; the extension exists because the complements have equal size. The counting at its core is the pigeonhole principle.

Theorem 3 (the Second Law is a theorem). Some reversible dynamics reproduces the process from A to B if and only if $W(A) \le W(B)$ (`second_law_is_a_theorem`). The right side mentions only the two macrostates.

Theorem 4 (no return). If $W(A) < W(B)$, no reversible dynamics reproduces the process from B back to A (`no_return`). A rise is irreversible as a macroscopic process, though every microscopic step is reversible.

Theorem 5 (each direction exactly when isentropic). The process is reproducible under some dynamics and its return under some dynamics exactly when $W(A) = W(B)$ (`reversible_iff_isentropic`); the two dynamics need not be the same.

Theorem 6 (Loschmidt located). Run any re-

versible dynamics backwards. Its inverse returns every microstate it carried out of A back into A, and yet, whenever $W(A) < W(B)$, it does not carry the macrostate B back into A (`loschmidt_resolved`). Reversal undoes trajectories, not macrostates: the microstates that φ brought from A fill only part of B, and the rest of B is carried elsewhere. Loschmidt's real question remains why the returning subset of B is never what gets prepared; the theorem does not answer it, and the asymmetry of preparation, the act, does.

The six theorems, the self-recursion of Section 6 and the prime count of Section 7 are bound in one theorem with nothing before its colon (`second_law_of_thermodynamics`).

5. Existence is self-recursive

Let a macroscopic description M assign to each microstate what is registered of it. The record of a state is every state M cannot tell apart from it (`record`). Every state lies in its own record, and the record of any state inside a record is that same record (`existence_is_self_recursive`). A record registers itself: registering what was registered returns the registration. This is the self-recursion on which the series rests existence (Islam 2026k, 2026n), proved here on every description; it is elementary, and Theorems 1 to 6 do not depend on it, since there A is any set of microstates.

6. The prime count equilibrates at its widest shell

Take the nine states $2^a 3^b$ with a and b at most 2, and register only the count of prime factors, $\Omega = a + b$ (`omega`). The five macrostates (`shell`) have multiplicities 1, 2, 3, 2 and 1 (`shell_multiplicities`), checked by the kernel's decision procedure. By Theorem 3 the count can be raised reproducibly from 0 to 1 and from 1 to 2, and it can never be lowered from 1 to 0, nor carried from the middle shell to $\Omega = 3$, by any reversible dynamics (`prime_arrow`). Since the middle shell is strictly the widest of the five, Theorem 3 gives it no reproducible exit to any other shell: whatever enters the widest shell stays there, the mark of equilibrium (Figure 1). Other moves are allowed, 0 to 3 and 4 to 0 among them, since only the multiplicities decide. The file encodes the states as the nine elements of a finite type with Ω read off the encoding; by unique factorization each pair (a, b) is one integer.

7. Scope, and what the act supplies

The theorem assumes two things about nature, and both are standard. Phase space is read at a finite resolution, so counts stand in for Liouville measure; the continuum statement replaces cardinality by measure and permutations by measure-preserving bijections; Theorem 1 carries over directly, and the converse does not without further hypotheses, and fails under energy-conserving flows between energy shells. The microscopic dynamics is reversible. Within those, the one hypothesis is the reproducibility of the process; the theorem then holds on every state space, for every dynamics, for every pair of macrostates. The entropy is Boltzmann's, of the macrostate; the paper claims nothing about Gibbs or von Neumann entropy.

The arrow itself enters through the act: a process is something prepared and repeated. The experimenter who prepares A and finds B, every time, has delivered reproducibility, and the theorem then fixes that W did not fall. That is the series' form: the form by theorem, the value by the act (Islam 2026m, 2026n). Here the form is the whole law, and the act is the existence of a repeatable preparation.

8. Prior work, and what is new

Table 1 places the paper. Jaynes (1965) is its foundation: his reproducibility argument is Theorem 1, and this paper owes him the form of the law. The resource theory of noisy operations characterizes which transformations a reversible process with a maximally mixed environment can achieve (Horodecki, Horodecki and Oppenheim 2003; Gour et al. 2015); in its microcanonical case it already contains the counting fact behind Theorems 2 and 3, in a probabilistic and operational setting, and so precedes this paper on that point. Lieb and Yngvason (1999) proved the entropy principle equivalent to their axioms on adiabatic accessibility, an order on states that sets aside the course of the process; this paper works on the dynamics itself.

To our knowledge, after the search reported here, what is new is the formalization and the framing: Jaynes's form of the Second Law and its converse, proved in Lean against Mathlib with every cone pinned; Loschmidt's objection located by a theorem that says what reversal does and what it cannot do; and the prime count as a worked instance whose widest shell has no reproducible exit. The two-way counting itself is elementary and is contained, in the microcanonical case, in the noisy-operations literature.

9. Falsifiers

Three, each forced by a claim of the paper; the compile is the standing falsifier of every cone.

F-Count. A miscount of the shells: any multiplicity other than 1, 2, 3, 2, 1 for $\Omega = 0$ to 4 on the nine states. The kernel's decision procedure checks it, so the aim is the encoding: show that state i of the file is not $2^{\lfloor i/3 \rfloor} 3^{i \bmod 3}$.

F-Process. One macroscopic process, prepared in A and found in B every time, with $W(B) < W(A)$ at a resolution fine enough that the microdynamics is a

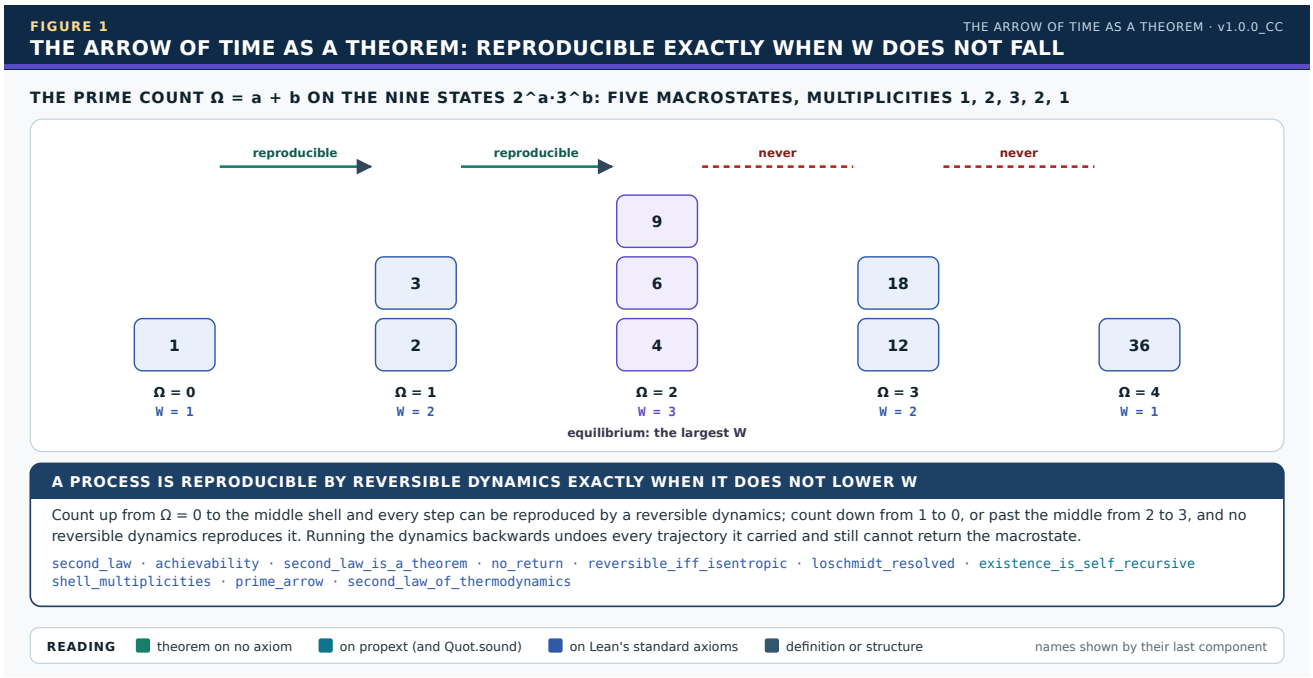


Figure 1 | The Second Law as a theorem, on the prime count. Nine states $2^a \cdot 3^b$ graded by $\Omega = a + b$ form five macrostates of multiplicity 1, 2, 3, 2, 1. By Theorem 3 the count is raised reproducibly from 0 to 1 and 1 to 2, never lowered from 1 to 0, and the widest shell has no reproducible exit.

Work	Position	This paper	Relation
Clausius 1865; Kelvin 1851	heat does not pass by itself from cold to hot; no engine turns heat wholly into work	the law as a characterization of possible processes, proved	extends
Boltzmann 1872	entropy rises, on molecular chaos	the rise from reversibility and reproducibility alone	extends
Loschmidt 1876; Zermelo 1896	reversible, recurrent mechanics cannot yield the law	reversal undoes trajectories and not the macrostate return (`loschmidt_resolved`)	extends
Carathéodory 1909; Lieb and Yngvason 1999	entropy from axioms on adiabatic accessibility	the law on the dynamics, both directions	adjacent
Jaynes 1965	a reproducible process cannot lower phase volume	Theorem 1, machine-checked; its converse, no return, isentropy and Loschmidt located	extends
Horodecki, Horodecki and Oppenheim 2003; Gour et al. 2015	noisy operations; microcanonical transformations	the same counting, here as deterministic reproducibility of macrostates	precedes
Lebowitz 1993; Albert 2000	Boltzmann entropy and the low-entropy past	the arrow carried by the repeatable preparation	adjacent
Jarzynski 1997; Crooks 1999	exact identities from a canonical initial state	a different premise and a different entropy	adjacent
Lean formalization, 2025 to 2026	ideal gas in PhysLean; Lieb–Yngvason in progress	the Second Law as a machine-checked characterization	extends

Table 1 | Positioning. Extends: *carried further;* corroborates: *agrees independently;* adjacent: *a different premise or domain, no formal engagement;* precedes: *holds the result first. No prior position is contradicted.*

permutation. It refutes the reversibility of the microdynamics at that resolution, since Theorem 1 leaves no other way.

F-Continuum. A measure-preserving bijection of a continuous phase space carrying a set of positive measure into a set of smaller measure. It cannot exist, which is the continuum form of Theorem 1; the aim is the passage from counts to measures, where the finite theorem is read.

10. Conclusion

The arrow of time asked how a two-way mechanics yields a one-way law. The answer proved here: the dynamics stays two-way, the law is a theorem about reproducible processes, and the direction enters with the preparation. The Second Law says which processes nature allows: those that do not lower the entropy. On every finite reversible dynamics, given reproducibility, that is a theorem, with its converse. A reproducible process never lowers the Boltzmann multiplicity, and every pair of macrostates that does not lower it is reproduced by some permutation. After a rise there is no return; each direction is reproducible

exactly when the multiplicity is unchanged; and the inverse dynamics, exact on every trajectory, cannot return the macrostate, which locates Loschmidt's objection in the asymmetry of preparation. Existence, the self-recursion of the record, is proved on every description. On the prime count the multiplicities are 1, 2, 3, 2, 1, and the widest shell has no reproducible exit.

The verdict, in the words of the reading guide, unchanged:

> Reversible mechanics obeys the Second Law. On every finite state space with reversible dynamics, a reproducible macroscopic process never lowers the Boltzmann multiplicity, and every pair of macrostates that does not lower it is reproduced by some permutation of the state space. Time-symmetric dynamics yields a one-way law: after a rise the return is never reproducible, each direction is reproducible exactly when the multiplicity is unchanged, and the inverse dynamics undoes every trajectory and never the rise. The arrow enters through the one hypothesis, reproducibility, delivered by the act of repeatable preparation. Existence is self-recursive. On the prime count the multiplicities are 1, 2, 3, 2, 1, and the widest shell has no reproducible exit. The arrow of time, as the Second Law states it, is a theorem.

Appendix A · The proof in Lean 4

The paper's file, `Second_Law_Theorem_v1_0_0_CC.lean`, printed in full and numbered by line: its header with the design and non-negotiables; the multiplicity and reproducibility; Theorems 1 to 6; the record and its self-recursion; the prime count; the binding theorem; and at its foot every pinned cone. It compiles on Lean 4.19.0 against Mathlib v4.19.0 with exit 0 and no message. Its SHA-256 is `cb6c4a320158d9b1a30afa52b478c0a704ccbf35e7bc96540f48fbd64b0287b5`, and the Markdown master of this paper carries the same file byte for byte in a machine-readable block at its foot.

```
/-
  Second_Law_Theorem_v1_0_0_CC.lean · the arrow of time as a theorem: the Second Law of Thermodynamics on reversible dynamics
  Claude-coded (_CC), 5 November 2026. Lean 4.19.0, Mathlib v4.19.0. No axiom declared, no sorry, no native_decide.

  DESIGN AND NON-NEGOTIABLES (read before editing).
  · The state space is finite and the microdynamics is a permutation of it: reversible, information-preserving,
    Liouville's theorem in its finite form. Nothing else about mechanics is assumed.
  · A macrostate is a finite set of microstates; its Boltzmann multiplicity W is its cardinality, and k log W its
    entropy. A process A → B is REPRODUCIBLE under a dynamics when every microstate prepared in A lands in B: the
    record "prepared in A" persists into the record "found in B". This is existence, stated on the prepared record,
    as Jaynes (1965) stated reproducibility. It is NOT required on microstates outside the prepared record: requiring
    it everywhere forbids every arrow (a referee's counterexample to an earlier draft of this programme).
  · Entropy here is Boltzmann's, of the macrostate. The theorems below are exactly about W; no claim is made about
    Gibbs or quantum entropy.

  What is proved:
    T.1  THE SECOND LAW: a reproducible process never lowers W (Jaynes's argument, machine-checked).
    T.2  ACHIEVABILITY: every pair of macrostates that does not lower W is reproduced by some permutation, chosen after
         the pair; physical dynamics with conservation laws can only restrict this.
    T.3  THE SECOND LAW IS A THEOREM: reproducible by reversible dynamics exactly when W does not fall.
    T.4  NO RETURN: if W rises, no reversible dynamics reproduces the return.
    T.5  REVERSIBLE EXACTLY WHEN ISENTROPIC: both directions reproducible exactly when W is equal.
    T.6  LOSCHMIDT LOCATED: the inverse dynamics undoes every trajectory it is given and still does not reproduce
         the return of the macrostate, whenever W rose. Why the returning states are never prepared is the
         asymmetry of preparation, outside this file.
    T.7  EXISTENCE IS SELF-RECURSIVE: every state is registered in its own record, and the record of anything in a
         record is that record.
    T.8  THE PRIME INSTANCE: nine states 2^a·3^b graded by Ω = a + b have multiplicities 1, 2, 3, 2, 1; the count can be
         raised from 0 to 1 and from 1 to 2 reproducibly, never lowered from 1 to 0, and never raised from 2 to 3.
         The states are encoded as Fin 9; by unique factorization each pair (a, b) is one integer.
    Ω    One theorem with nothing before its colon.
  Every theorem is pinned by #guard_msgs at the foot; a drift fails the compile.
-/
import Mathlib.Logic.Equiv.Fintype
import Mathlib.Data.Fintype.EquivFin
import Mathlib.Data.Finset.Card
import Mathlib.Data.Fintype.Card

namespace SecondLawTheorem

open Finset

variable {α : Type*} [Fintype α] [DecidableEq α]

/-- THE BOLTZMANN MULTIPLICITY of a macrostate: the number of its microstates. -/
def W (A : Finset α) : ℕ := A.card

/-- A REPRODUCIBLE PROCESS A → B under a reversible dynamics φ: every microstate prepared in A lands in B. -/
def Reproducible (φ : Equiv.Perm α) (A B : Finset α) : Prop := ∀ x ∈ A, φ x ∈ B

omit [Fintype α] [DecidableEq α] in
/-- T.1 · THE SECOND LAW: a reproducible process never lowers the Boltzmann multiplicity. -/
theorem second_law (φ : Equiv.Perm α) (A B : Finset α) (h : Reproducible φ A B) : W A ≤ W B :=
  Finset.card_le_card_of_injOn φ h φ.injective.injOn

/-- T.2 · ACHIEVABILITY: a process that does not lower the multiplicity is reproducible under some reversible
    dynamics. Choose a part of B as large as A, match A to it one to one, and extend the matching to a permutation of
    the whole space. -/
theorem achievability (A B : Finset α) (h : W A ≤ W B) : ∃ φ : Equiv.Perm α, Reproducible φ A B := by
  obtain ⟨t, htB, htc⟩ := Finset.exists_subset_card_eq (s := B) (show A.card ≤ B.card from h)
  have hc : Fintype.card { x // x ∈ A } = Fintype.card { x // x ∈ t } := by
    rw [Fintype.card_coe, Fintype.card_coe, htc]
  let e : { x // x ∈ A } ≃ { x // x ∈ t } := Fintype.equivOfCardEq hc
  refine ⟨Equiv.extendSubtype e, fun x hx => ?_⟩
  rw [Equiv.extendSubtype_apply_of_mem e x hx]
  exact htB (e ⟨x, hx⟩).2

/-- T.3 · THE SECOND LAW IS A THEOREM: a macroscopic process is reproducible by reversible dynamics exactly when it
    does not lower the Boltzmann multiplicity. -/
theorem second_law_is_a_theorem (A B : Finset α) :
    (∃ φ : Equiv.Perm α, Reproducible φ A B) ↔ W A ≤ W B :=
  ⟨fun ⟨φ, h⟩ => second_law φ A B h, achievability A B⟩

omit [Fintype α] [DecidableEq α] in
/-- T.4 · NO RETURN: when the multiplicity rises, no reversible dynamics reproduces the way back. -/
theorem no_return (A B : Finset α) (h : W A < W B) : ¬ ∃ ψ : Equiv.Perm α, Reproducible ψ B A :=
  fun ⟨ψ, hψ⟩ => Nat.lt_irrefl _ (Nat.lt_of_lt_of_le h (second_law ψ B A hψ))

/-- T.5 · REVERSIBLE EXACTLY WHEN ISENTROPIC: a process and its return are both reproducible exactly when the
    multiplicities are equal. -/
theorem reversible_iff_isentropic (A B : Finset α) :
    ((∃ φ : Equiv.Perm α, Reproducible φ A B) ∧ (∃ ψ : Equiv.Perm α, Reproducible ψ B A)) ↔ W A = W B := by
  rw [second_law_is_a_theorem, second_law_is_a_theorem]
  exact ⟨fun ⟨h1, h2⟩ => Nat.le_antisymm h1 h2, fun h => ⟨h.le, h.ge⟩⟩

omit [Fintype α] in
/-- T.6 · LOSCHMIDT LOCATED: run the dynamics backwards. It returns every microstate it carried, the image of A, to
    A; and when the multiplicity rose it does not carry the macrostate B back into A. Microscopic reversal is exact,
    macroscopic return is impossible. -/
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```
theorem loschmidt_resolved (φ : Equiv.Perm α) (A B : Finset α) (hlt : W A < W B) :
    Reproducible φ.symm (A.image φ) ∧ ¬ Reproducible φ.symm B A := by
  refine (fun x hx => ?_, fun hb => no_return A B hlt (φ.symm, hb))
  obtain (y, hy, rfl) := Finset.mem_image.mp hx
  simpa using hy

/-- The record of a state under a macroscopic description M: every state M cannot tell apart from it. -/
def record {ι : Type*} [DecidableEq ι] (M : α → ι) (x : α) : Finset α := univ.filter (fun y => M y = M x)

omit [DecidableEq α] in
/-- T.7 · EXISTENCE IS SELF-RECURSIVE: every state is registered in its own record, and the record of any state in a
    record is that same record. -/
theorem existence_is_self_recursive {ι : Type*} [DecidableEq ι] (M : α → ι) (x : α) :
    x ∈ record M x ∧ ∀ y ∈ record M x, record M y = record M x := by
  refine (by simp [record], fun y hy => ?_)
  have hy' : M y = M x := (Finset.mem_filter.mp hy).2
  ext z
  simp [record, hy']

/-! ## T.8 · The prime instance: nine states 2^a·3^b, graded by the count of prime factors -/

/-- The state i of Fin 9 is 2^a·3^b with a = i / 3 and b = i % 3. -/
def omega (i : Fin 9) : ℕ := i.val / 3 + i.val % 3

/-- The macrostate of count k: every state with Ω = k. -/
def shell (k : ℕ) : Finset (Fin 9) := univ.filter (fun i => omega i = k)

/-- THE MULTIPLICITIES OF THE COUNT: 1, 2, 3, 2, 1 for Ω = 0, 1, 2, 3, 4. -/
theorem shell_multiplicities :
    W (shell 0) = 1 ∧ W (shell 1) = 2 ∧ W (shell 2) = 3 ∧ W (shell 3) = 2 ∧ W (shell 4) = 1 := by
  decide

/-- THE PRIME ARROW: the count can be raised from 0 to 1 and from 1 to 2 reproducibly; it can never be lowered from 1
    to 0, and never raised from 2 to 3, by any reversible dynamics. -/
theorem prime_arrow :
    (∃ φ : Equiv.Perm (Fin 9), Reproducible φ (shell 0) (shell 1)) ∧
    (∃ φ : Equiv.Perm (Fin 9), Reproducible φ (shell 1) (shell 2)) ∧
    (¬ ∃ φ : Equiv.Perm (Fin 9), Reproducible φ (shell 1) (shell 0)) ∧
    (¬ ∃ φ : Equiv.Perm (Fin 9), Reproducible φ (shell 2) (shell 3)) := by
  obtain (h0, h1, h2, h3, -) := shell_multiplicities
  refine (achievability _ _ ?_, achievability _ _ ?_, no_return _ _ ?_, no_return _ _ ?_) <;> omega

/-! ## Ω · The Second Law of Thermodynamics, as a theorem -/

/-- THE SECOND LAW OF THERMODYNAMICS AS A THEOREM, nothing before the colon: on every finite state space with
    reversible dynamics, a reproducible process never lowers the Boltzmann multiplicity, every process that does not
    lower it is reproducible, the two are one statement, the return of a rise is never reproducible, a process is
    reversible exactly when it is isentropic, the inverse dynamics undoes trajectories and not the rise, existence is
    self-recursive, and on the prime count the multiplicities are 1, 2, 3, 2, 1 with the arrow they force. -/
theorem second_law_of_thermodynamics :
    type_of% @second_law ∧
    type_of% @achievability ∧
    type_of% @second_law_is_a_theorem ∧
    type_of% @no_return ∧
    type_of% @reversible_iff_isentropic ∧
    type_of% @loschmidt_resolved ∧
    type_of% @existence_is_self_recursive ∧
    type_of% @shell_multiplicities ∧
    type_of% @prime_arrow :=
  (@second_law, @achievability, @second_law_is_a_theorem, @no_return, @reversible_iff_isentropic,
   @loschmidt_resolved, @existence_is_self_recursive, @shell_multiplicities, @prime_arrow)

end SecondLawTheorem

/-! ## Cones, pinned -/

/-- info: 'SecondLawTheorem.second_law' depends on axioms: [propext, Classical.choice, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.second_law
/-- info: 'SecondLawTheorem.achievability' depends on axioms: [propext, Classical.choice, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.achievability
/-- info: 'SecondLawTheorem.second_law_is_a_theorem' depends on axioms: [propext, Classical.choice, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.second_law_is_a_theorem
/-- info: 'SecondLawTheorem.no_return' depends on axioms: [propext, Classical.choice, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.no_return
/-- info: 'SecondLawTheorem.reversible_iff_isentropic' depends on axioms: [propext, Classical.choice, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.reversible_iff_isentropic
/-- info: 'SecondLawTheorem.loschmidt_resolved' depends on axioms: [propext, Classical.choice, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.loschmidt_resolved
/-- info: 'SecondLawTheorem.existence_is_self_recursive' depends on axioms: [propext, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.existence_is_self_recursive
/-- info: 'SecondLawTheorem.shell_multiplicities' depends on axioms: [propext, Classical.choice, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.shell_multiplicities
/-- info: 'SecondLawTheorem.prime_arrow' depends on axioms: [propext, Classical.choice, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.prime_arrow
/-- info: 'SecondLawTheorem.second_law_of_thermodynamics' depends on axioms: [propext, Classical.choice, Quot.sound] -/
#guard_msgs in #print axioms SecondLawTheorem.second_law_of_thermodynamics
```

Author's Provenance and Method Disclosure

This paper was developed under Trisduction, a verification and organizing discipline that adds the cited results no warrant; the method and its executable batteries are stated in full in the master reference (Islam 2026j), and the one-bit method is set out as a procedure in its companion volume (Islam 2026m).

Author and method. Mohammad F. Islam, PhD, independent researcher, sole author and sole authority for every claim and every error. The paper was directed by the author and drafted with Claude (Anthropic) as scribe: the model wrote and compiled the Lean file, drew the figure, ran the literature search and checked the citations against their sources, and argued against the author's positions on instruction during development; an independent review of an earlier draft found that its record-persistence premise, demanded on every microstate, admits no arrow, and this edition states reproducibility on the prepared record instead. The model holds no authorship. The thesis, that the Second Law is a theorem and existence its self-recursive ground, is the author's.

The register. In the programme's vocabulary: nine theorems stand at the grade of Lean's standard axioms and one on propext and Quot.sound; the repeatable preparation is delivered by the act; $\Delta M = 0$ on the cited physics.

Audit. This edition has not been prosecuted by the external substrates, and no audit result is claimed beyond the independent review reported above.

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